

Questions for: Patrick Parsons

Question 1: Remote work has become more popular over the past few years. We would like to look to see if there are any differences in the proportion of working moms who choose from the following options for working: part-time remote work, part-time in-person work, full-time remote work, and full-time in-person work. From our sample of 500 working moms, we found that 180 would prefer part-time remote work, 118 would prefer part-time in-person work, 95 would prefer full-time remote work, and the rest would prefer full-time in-person work.

- Conduct the appropriate test to determine if there are any differences in the proportions of working moms who prefer the different work options.
- Which, if any, follow-up confidence intervals would you want to create? Be sure to identify what type of interval, why that type, and which work options they would be created for. (You do not need to do any of the interval calculations).

Question 2: Let $Y_1, Y_2, \dots, Y_8$ be a random sample from a Normal population with mean 0 and variance 1. Given that U is chi-squared with 8 degrees of freedom, V is chi-squared with 6 degrees of freedom, $\bar{Y}$ is the sample mean of the 8 observations, and that Y_9 is a new observation from the sample population, find the distributions of the following (being sure you can show/explain)...

$$\frac{2Y_7}{\sqrt{U/2}}$$

$$\frac{4(8\bar{Y}^2 + Y_9^2)}{U}$$

Question 3: We would like to look at reading grade-level growth during 6th grade – specifically, checking to see if it has increased at the end of 6th grade. We took a random sample of 10 sixth graders and got the following:

Beginning	7.1	6.4	5.5	6.2	5.1	4.9	5.7	5.2	6.3	6.0
End	8.2	6.8	6.6	6.2	5.8	4.5	5.9	6.5	6.7	7.3

- Conduct the appropriate hypothesis test using a 5% significance level. Be sure to show all steps and be able to explain the test chosen.
- Would you want to create a confidence interval for the difference? Why or why not?